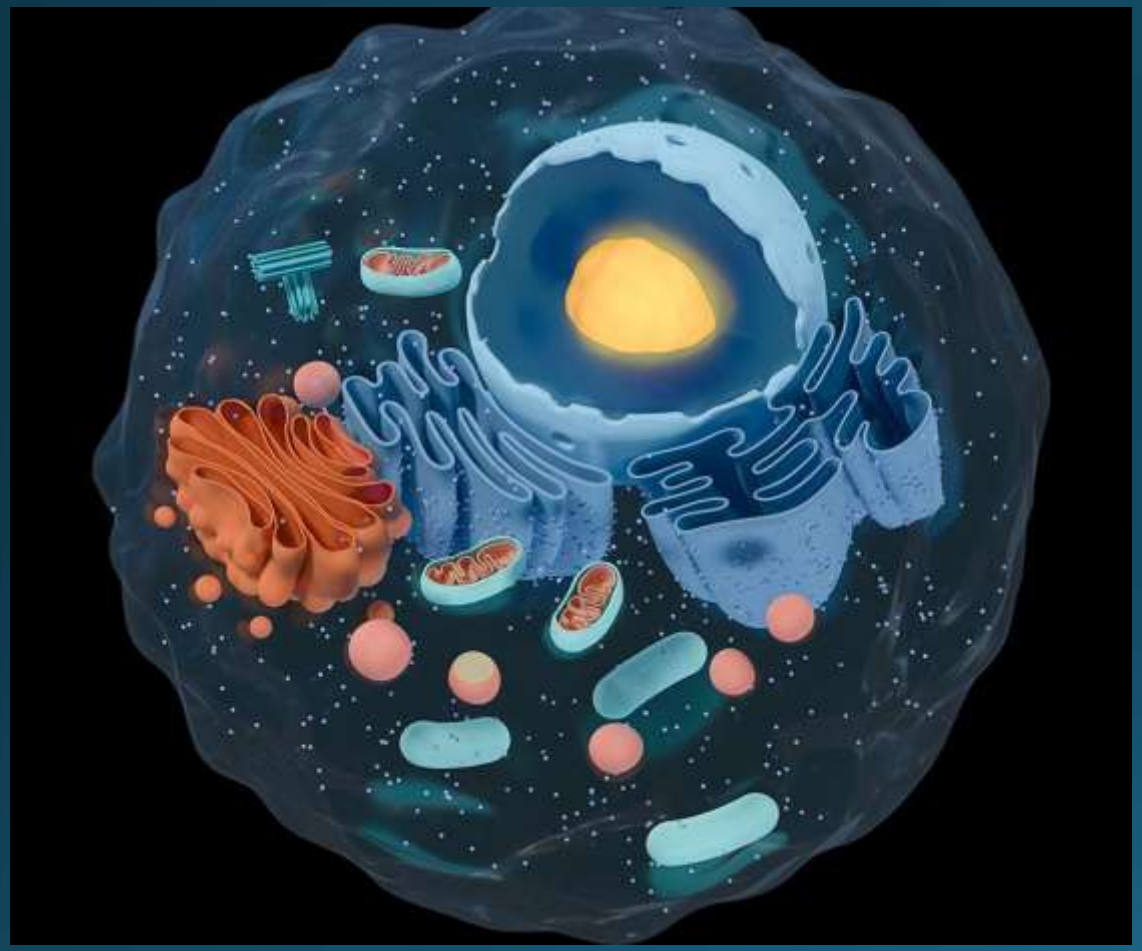


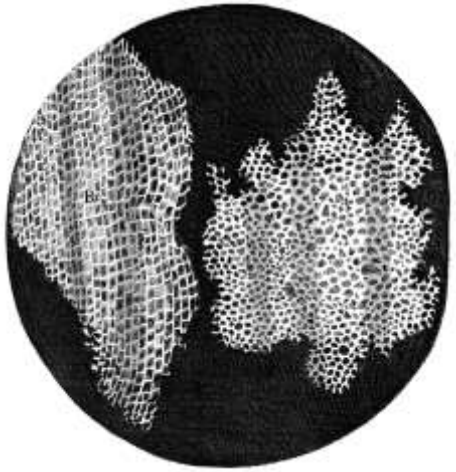
INTRODUCTION TO BIOLOGY
BIO-101

Cell Structure & Function
Lecture 03

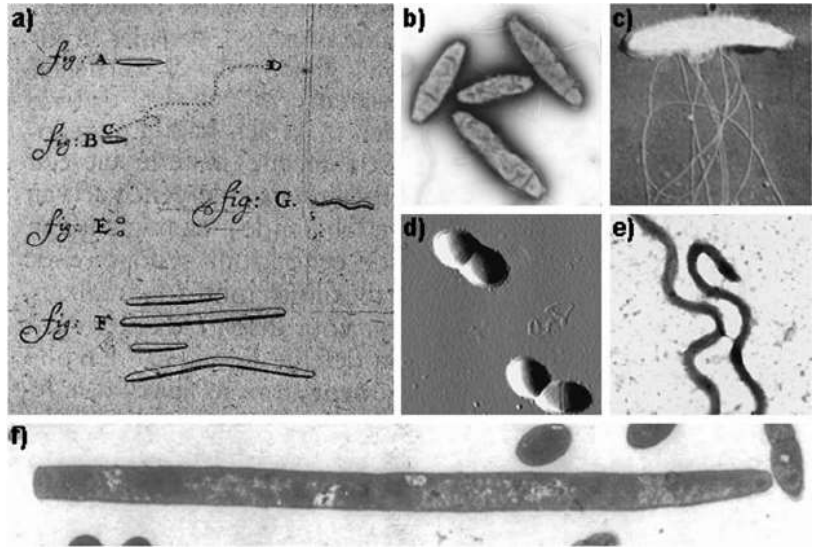
BRAC UNIVERSITY
Inspiring Excellence

Early Discoveries

- Mid 1600s - **Robert Hooke** observed and described cells in cork
- Late 1600s - **Antonie van Leeuwenhoek** observed microorganisms
- 1820s - **Robert Brown** observed and named nucleus in plant cells



Cells in cork



Bacteria under microscope

Cell division in plant cell

Developing Cell Theory

Cell Theory

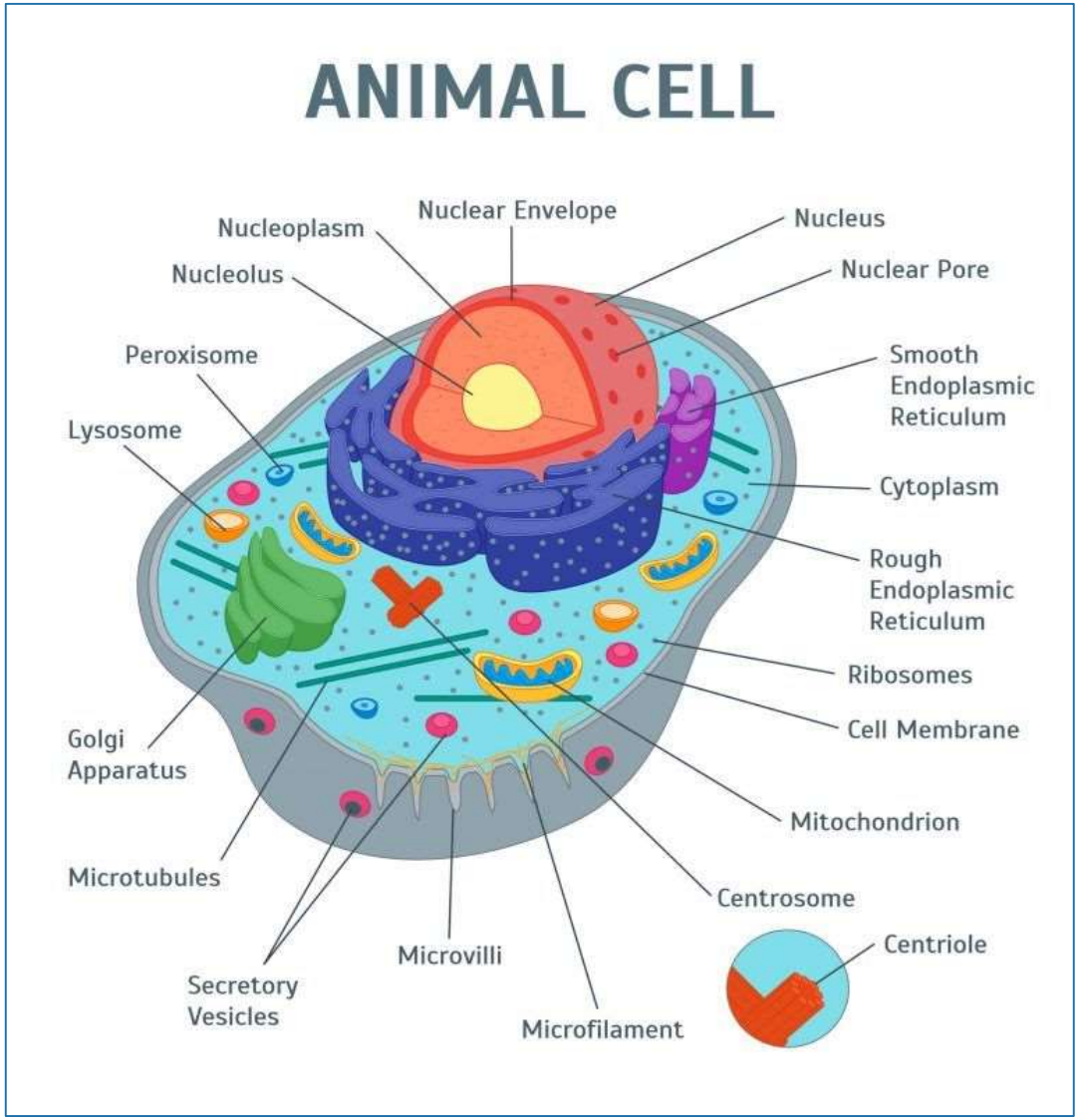
1) Every organism is composed of one or more cells

2) Cell is smallest unit having properties of life

3) Continuity of life arises from growth and division of single cells

Cell

- Smallest unit of life.
- Can survive on its own or has potential to do so.
- Is highly organized for metabolism.
- Senses and responds to environment.
- Has potential to reproduce.



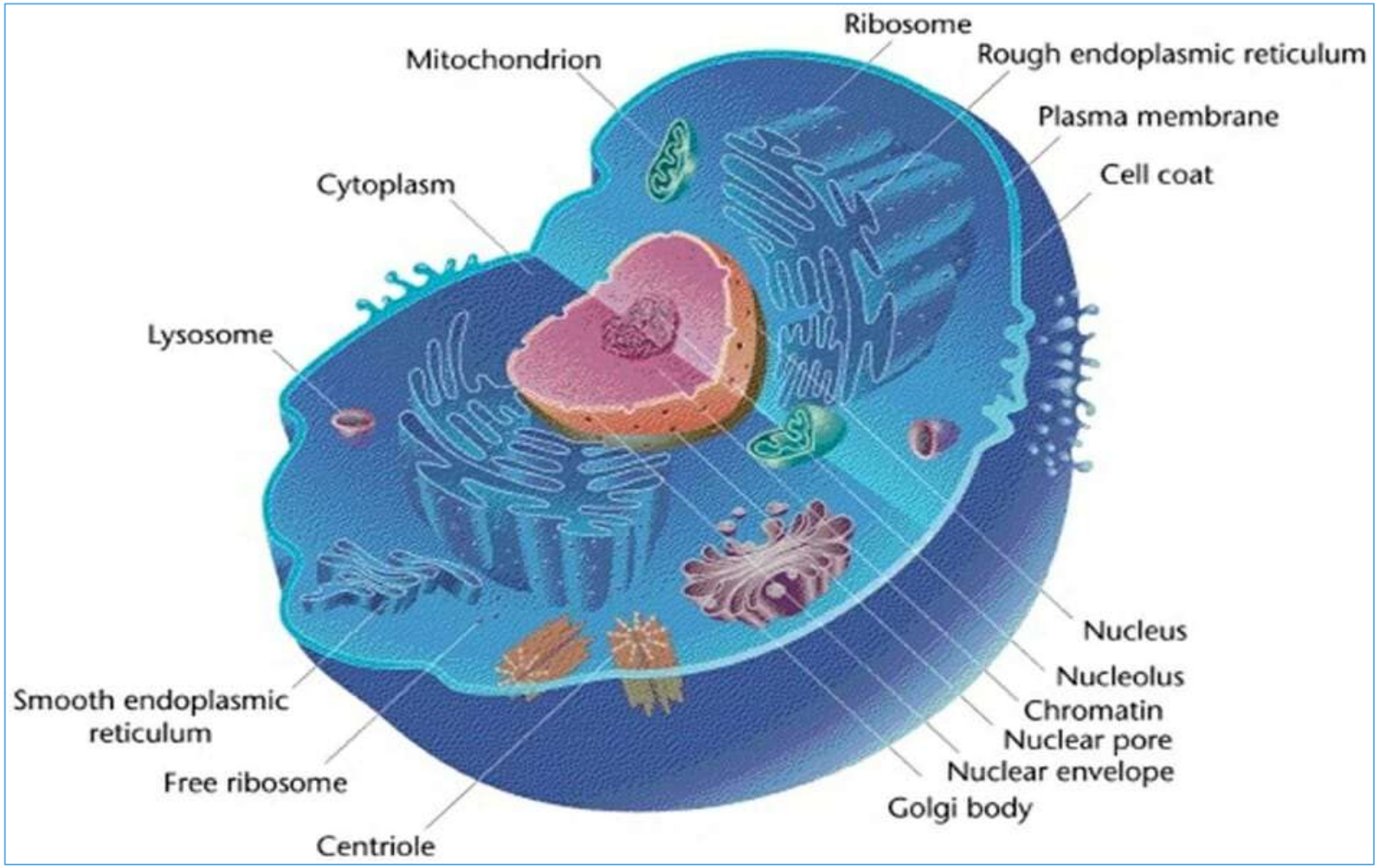
Structure of Cells

All start out life with:

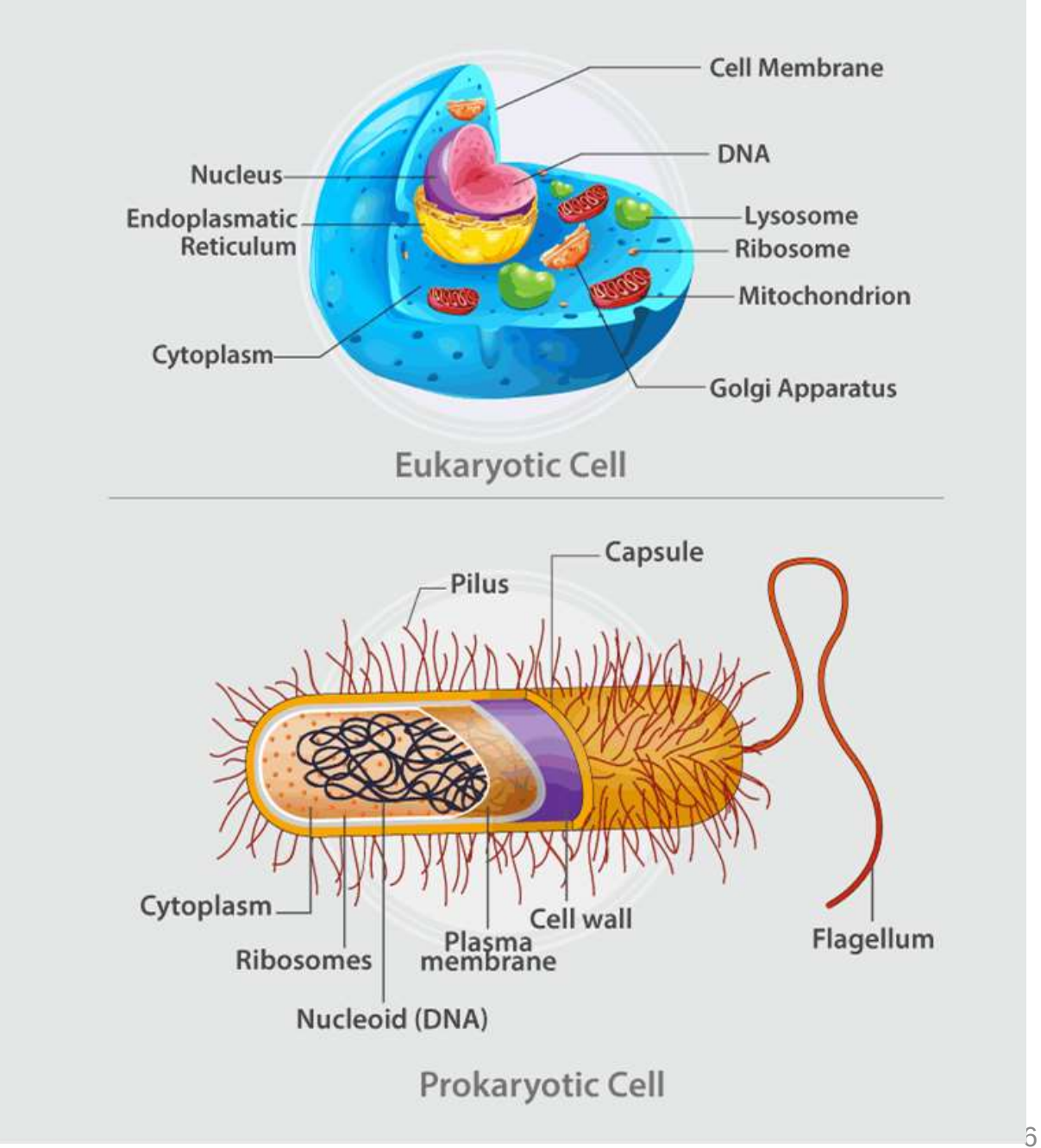
- Plasma membrane
- Nucleus/DNA
- Cytoplasm

Two types:

- Prokaryotic
- Eukaryotic



Prokaryotic Cell
&
eukaryotic cell



Prokaryotic and Eukaryotic Cells Venn Diagram

ScienceFacts.net

Prokaryotic Cells

- Small and simple
- 0.1 to 5.0 μm in size
- Unicellular
- Nucleus is absent
- Circular DNA
- Single haploid (n) chromosome
- Lack membrane-bound organelles
- Reproduce both sexually and asexually
- Cell division by binary fission
- Examples are bacteria and archaea cells

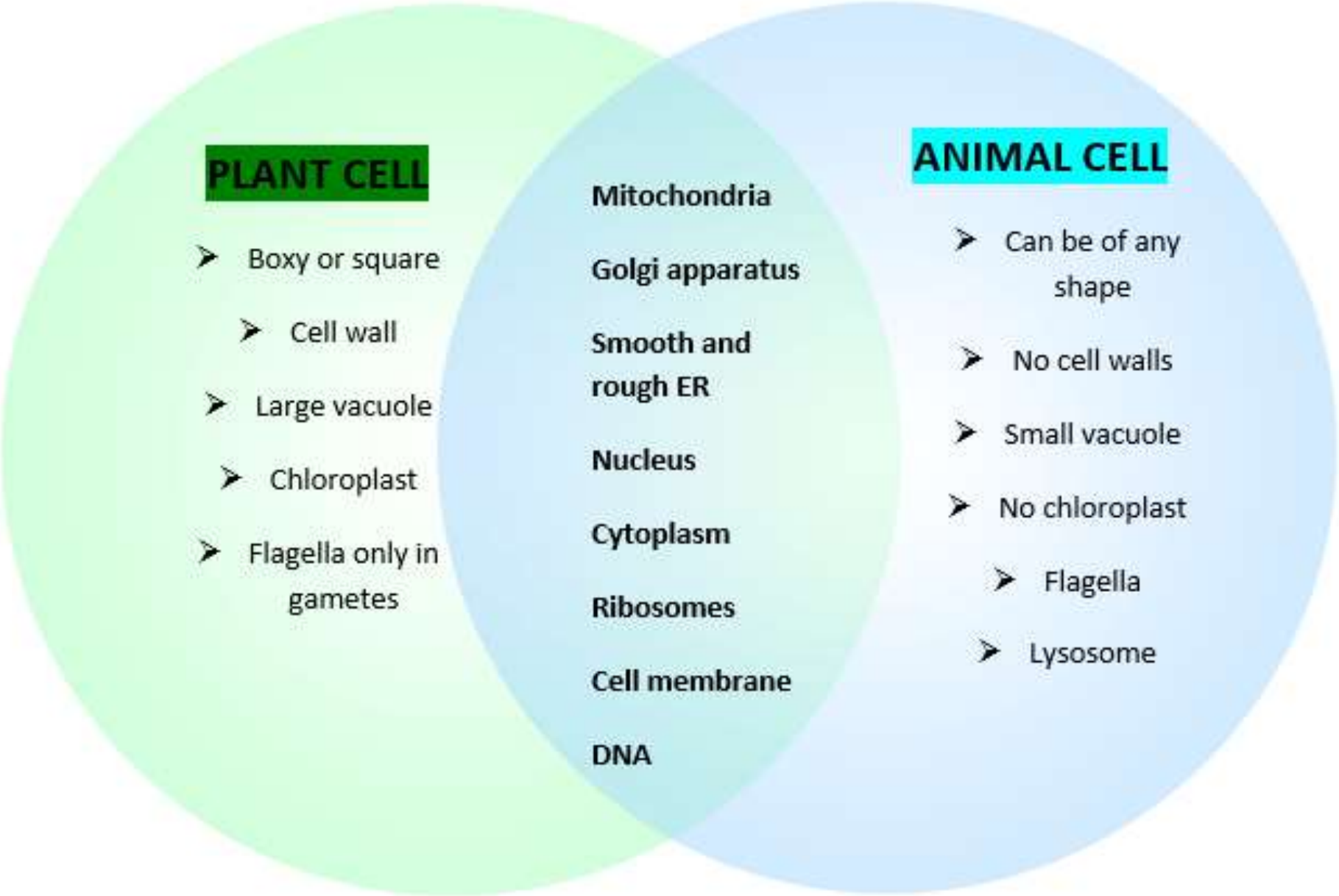
Similarities

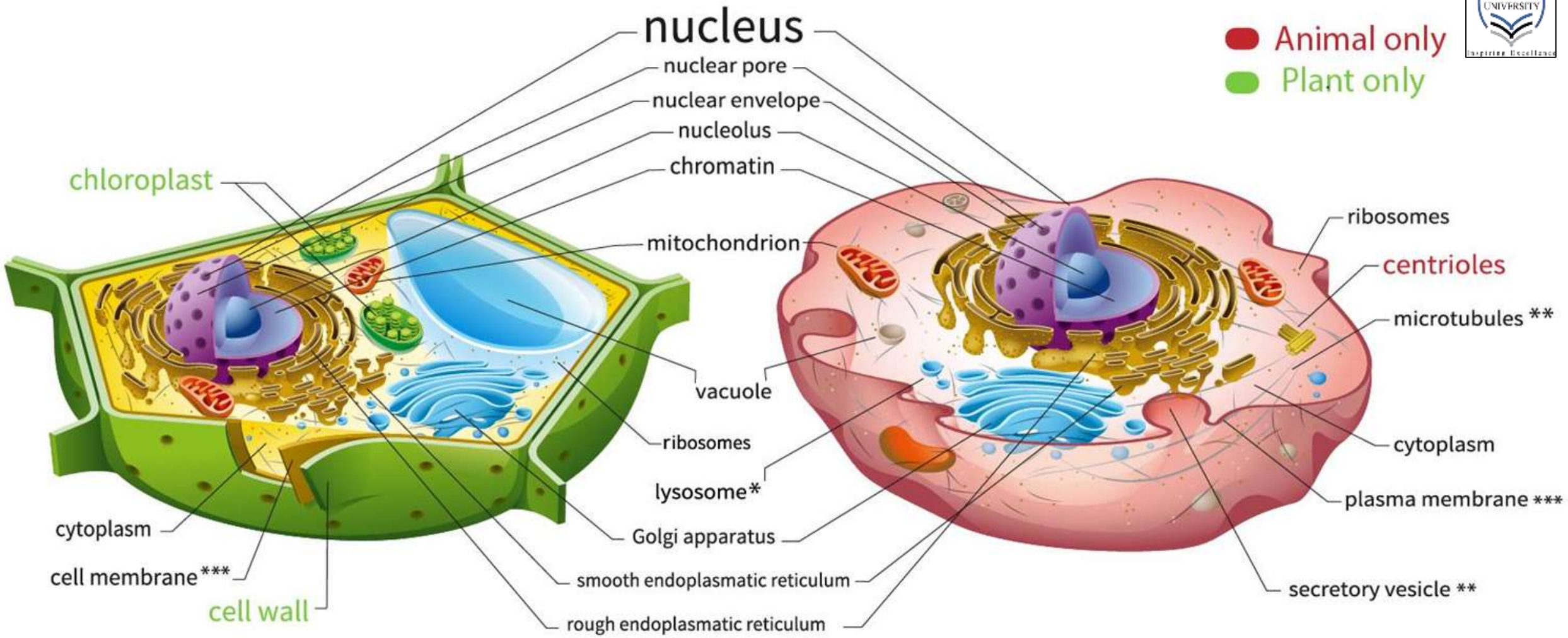
- Have cell (plasma) membrane
- Have cytoplasm
- Have ribosomes
- Have DNA

Eukaryotic Cells

- Large and complex
- 10 to 100 μm in size
- Unicellular or multicellular
- Nucleus is present
- Linear DNA
- Paired diploid (2n) chromosome
- Has membrane-bound organelles
- Mostly reproduce sexually
- Cell division by mitosis
- Examples are plant and animal cells, including humans

Animal Cell & Plant Cell





* Plants may have lytic vacuoles, which act like lysosomes in animal cells.

** Although they're not labelled here, plant cells have microtubules and secretory vesicles, too.

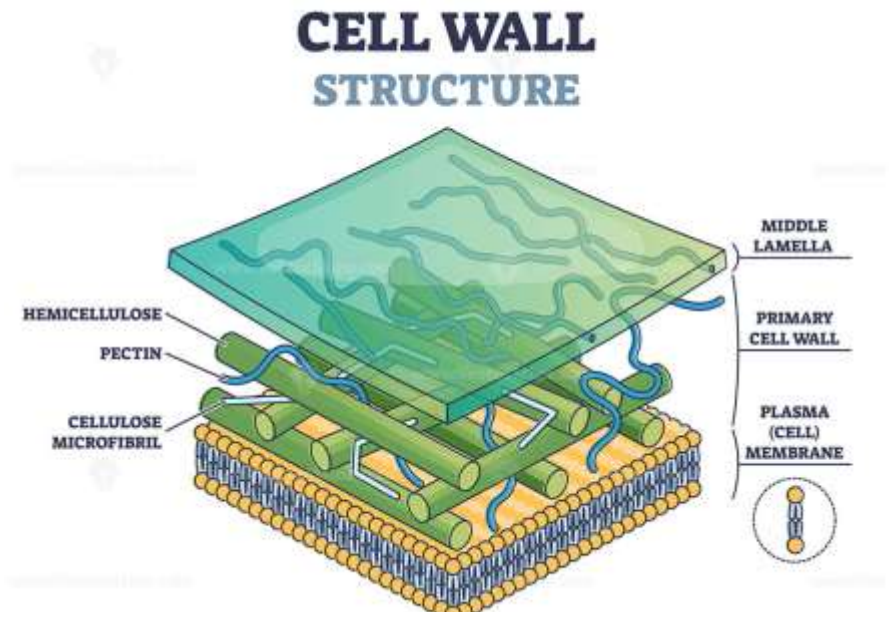
*** Cell membrane and plasma membrane are just different names for the same structure.

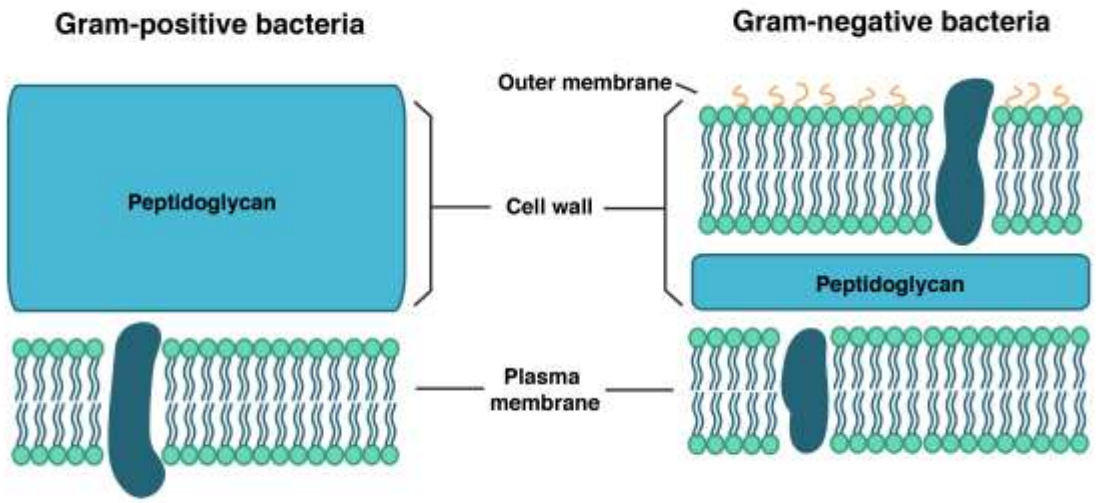
Structure of Cells

Cell wall

The cell wall is an integral component of many prokaryotic cell and plant cell. However it is absent in animal cells.

- The plant cell wall provides definite shape, strength, and rigidity
- It also provides protection against mechanical stress and physical shocks
- It helps to control cell expansion due to the intake of water
- It helps in preventing water loss from the cell
- It is responsible for transporting substances between and across the cell
- It acts as a barrier between the interior cellular component and the external environment





Structure of Cells
(Cell Membrane)

- The cell membrane, also called the plasma membrane, is found in all cells and separates the interior of the cell from the outside environment.
- The cell membrane consists of a lipid bilayer that is semipermeable.
- The cell membrane regulates the transport of materials entering and exiting the cell.

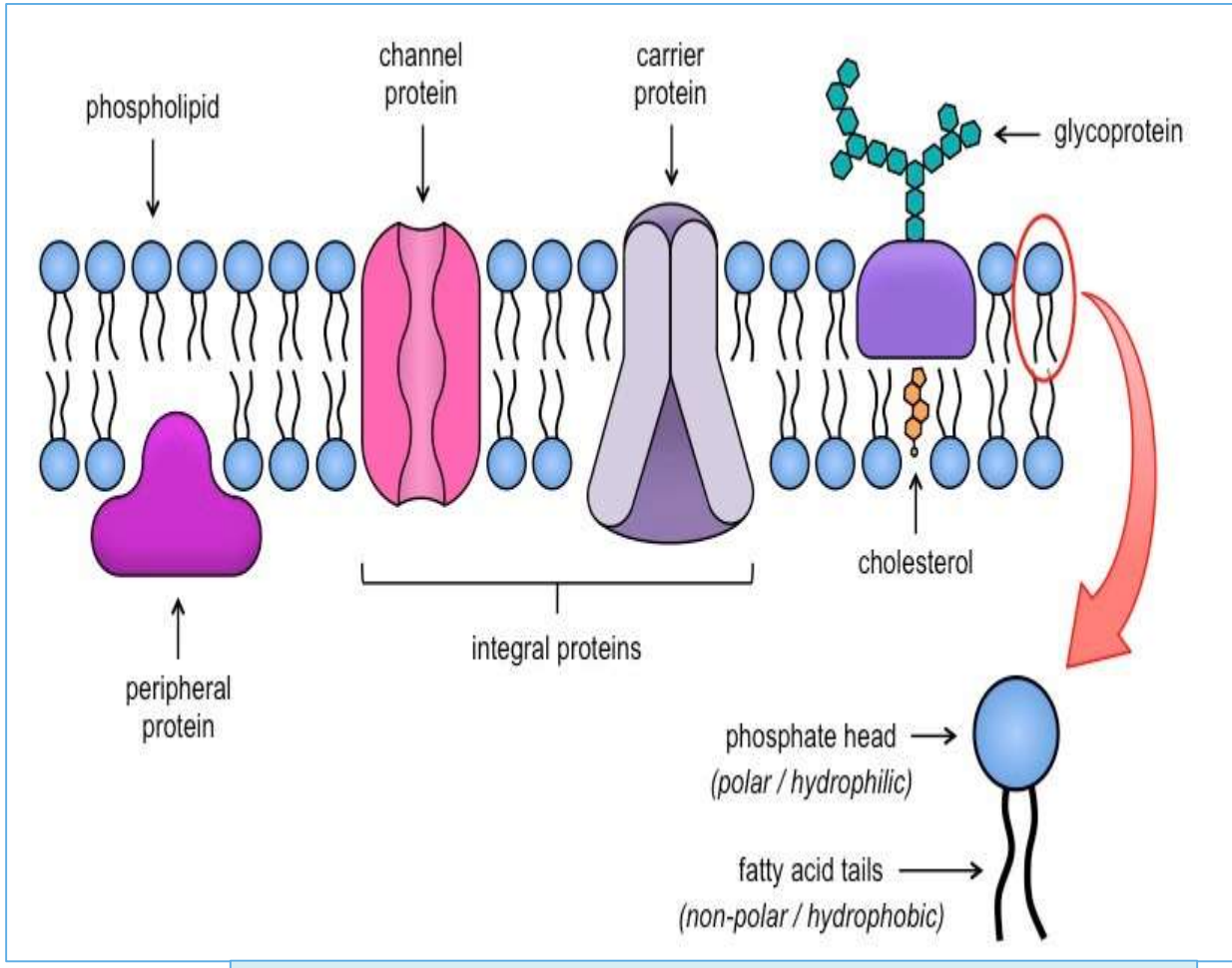
Components of Plasma Membrane

Components	Location
Phospholipid	The main fabric of plasma membrane
Cholesterol	Between phospholipids and phospholipid bilayers
Integral proteins	Embedded within phospholipid layers
Peripheral proteins	Inner or outer surface of the phospholipid bilayer
Carbohydrates	Attached to proteins on outside membrane layers

Structure of Cells
(Cell Membrane)

Fluid Mosaic Model

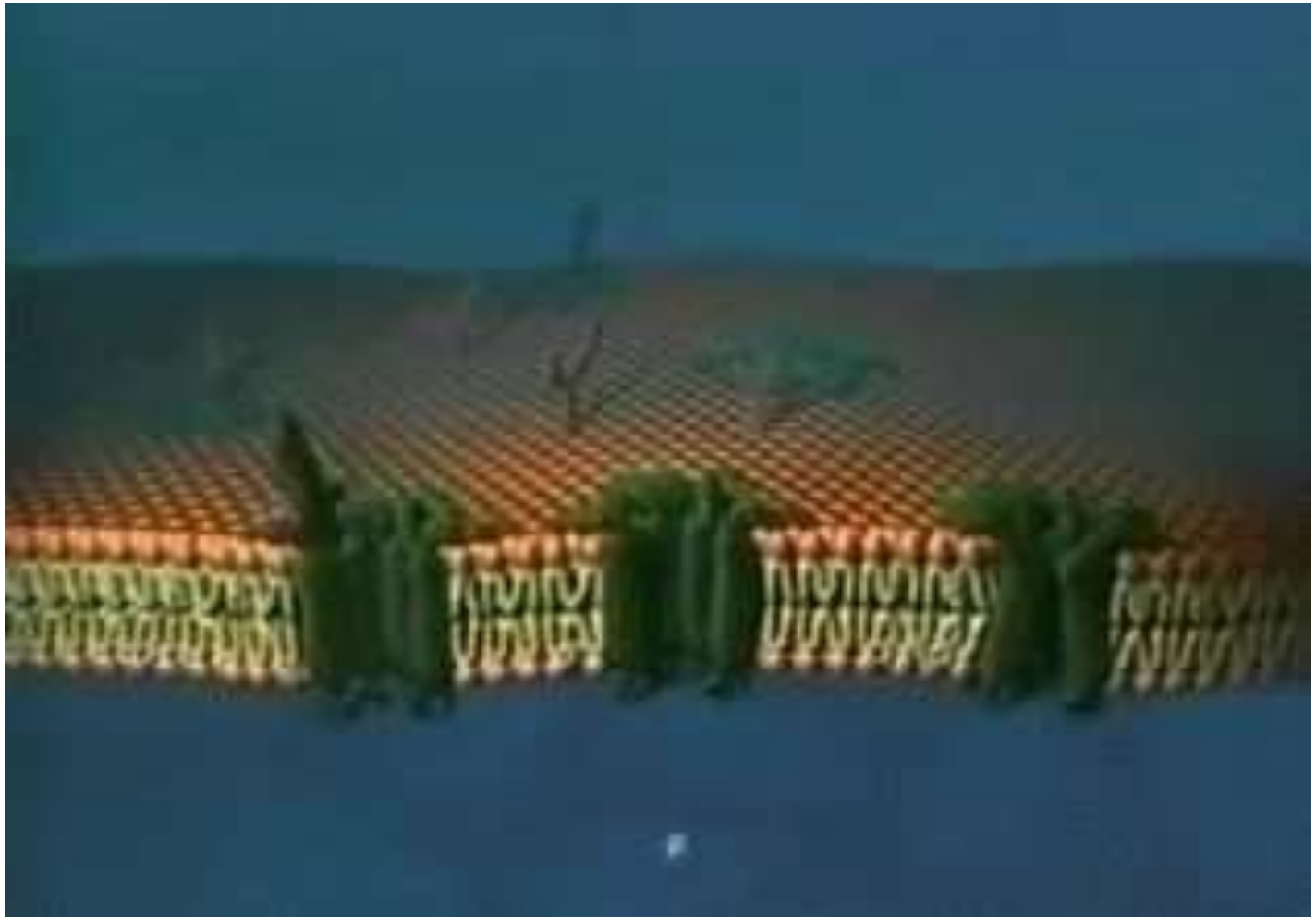
- The fluid mosaic model was proposed by S.J. Singer and Garth L. Nicolson.
- This model explains the structure of the plasma membrane of animal cells as a mosaic of components such as phospholipids, proteins, cholesterol, and carbohydrates.
- These components give a fluid character to the membranes.
- Each phospholipid has a hydrophilic head pointing outside and a hydrophobic tail forming the inside of the bilayer.
- Cholesterol and proteins are embedded in the bilayer that gives the membrane a mosaic look. Each component has a specific function to perform.



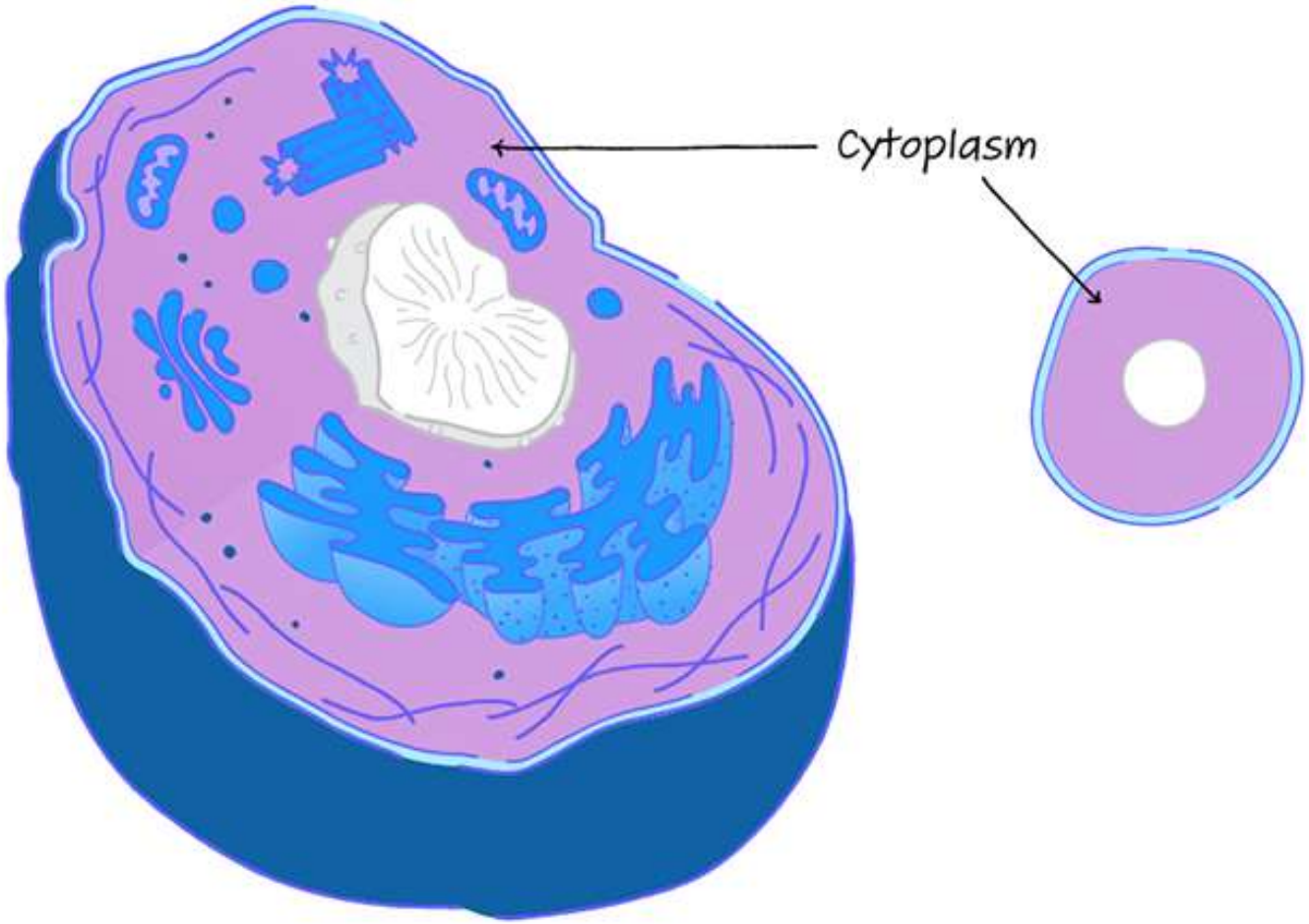
Cell Membrane (Fluid Mosaic Model)

Structure of Cells
(Cell Membrane)

Fluid Mosaic Model

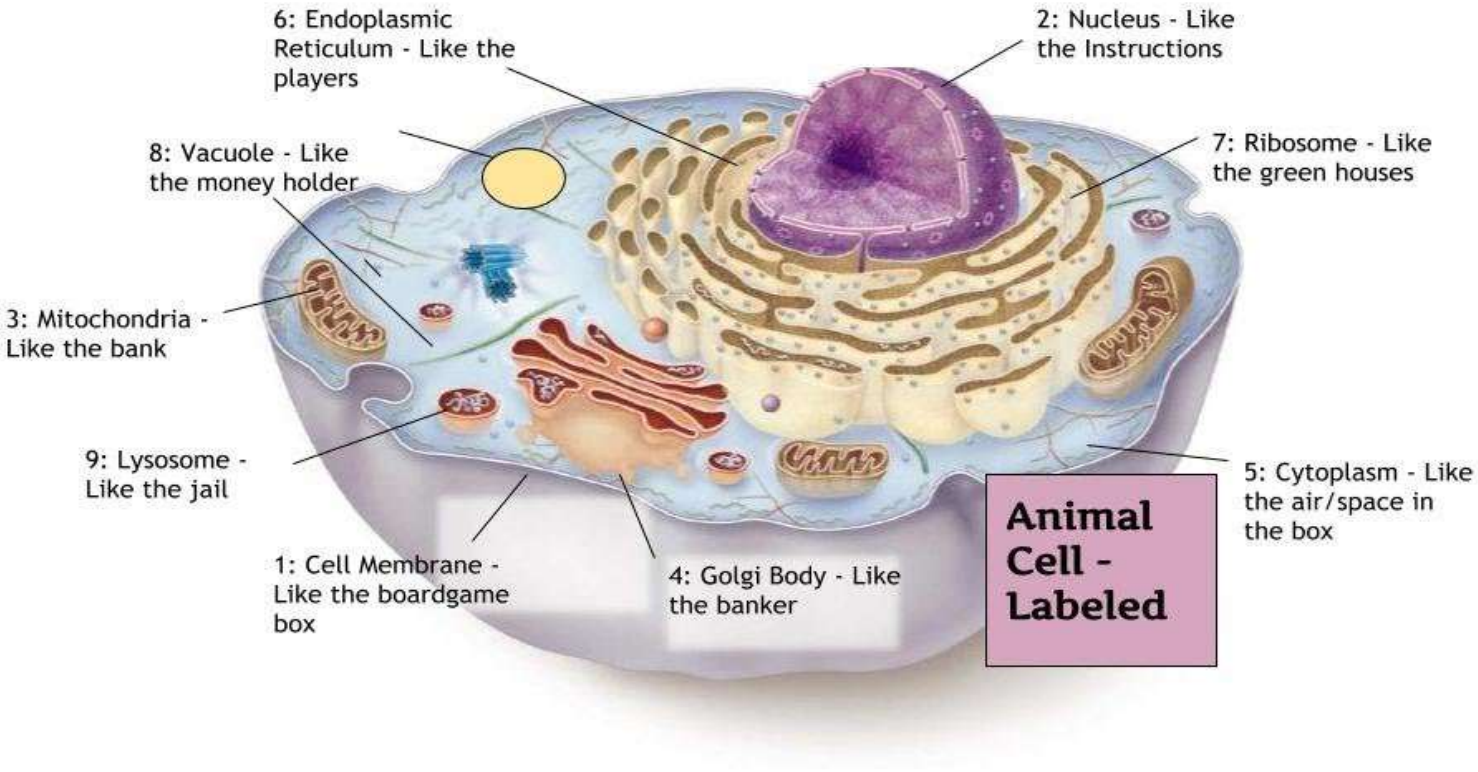


Structure of Cells
(Cytoplasm)



Structure of Cells
(Cytoplasm)

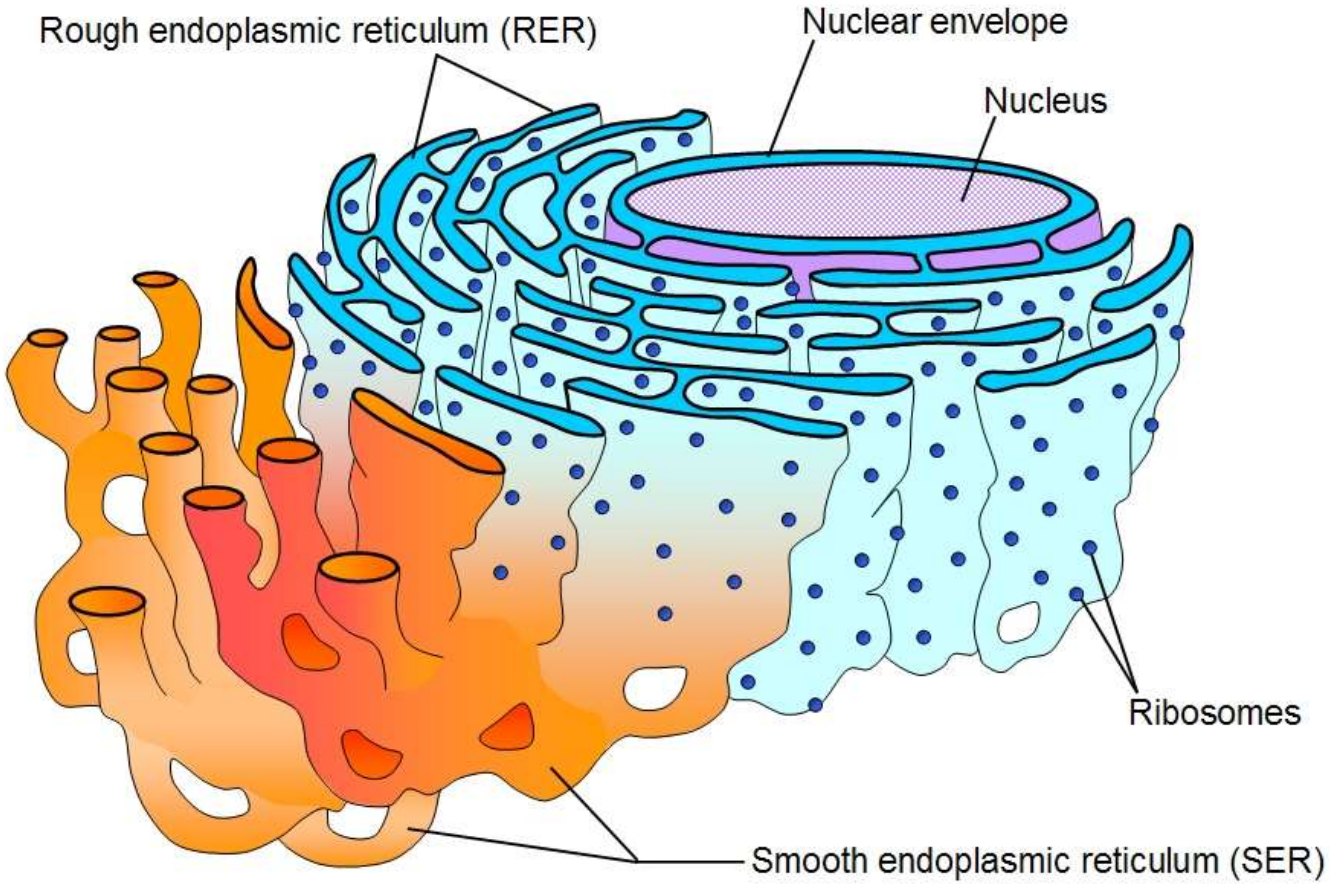
Endoplasmic reticulum (ER)
Golgi bodies
Mitochondria
Vesicles
Lysosomes (only in animal)
Plastids (only in plant)
Cytoskeleton



Structure of Cells
(Cytoplasm)

Endoplasmic Reticulum (ER)

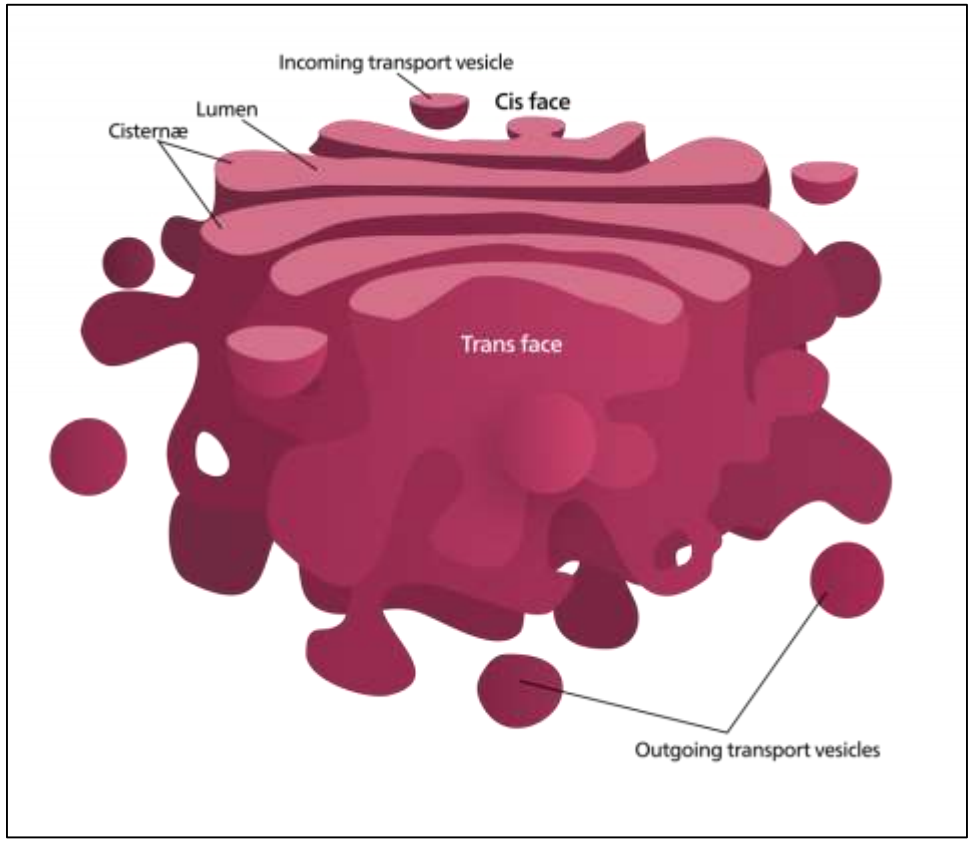
- In animal cells, continuous with nuclear membrane
- Extends throughout cytoplasm
- Two regions - rough and smooth
- Ribosomes are present on the rough ER
- Due to the absence of ribosomes the SER appears smooth



Structure of Cells (Cytoplasm)

Golgi Bodies

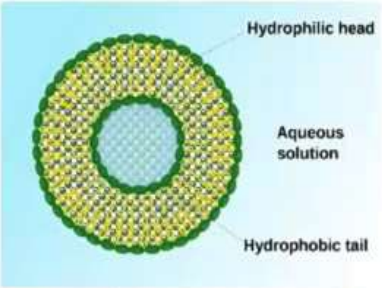
- Put finishing touches on proteins and lipids that arrive from ER
- Package finished material for shipment to final destinations
- Material arrives and leaves in vesicles



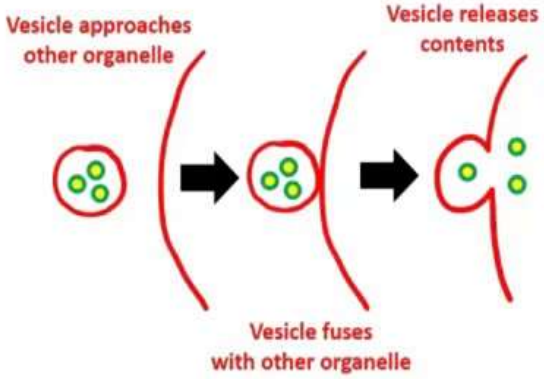
Structure of Cells (Cytoplasm)

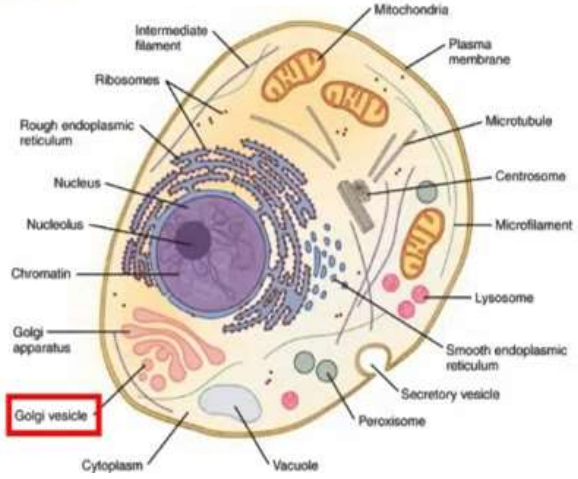
Vesicles

- Function: transport materials throughout cell
  - Like shipping boxes
- Structure: small spheres of membrane
  - Same material as RER, Golgi, plasma membrane, etc
    - Part of ENDOMEMBRANE system
    - Can fuse with or pinch off from any other membrane-bound organelle
- Found in ALL eukaryotic cells



https://commons.wikimedia.org/wiki/File:Liposome_scheme-en.svg





Structure of Cells (Cytoplasm)

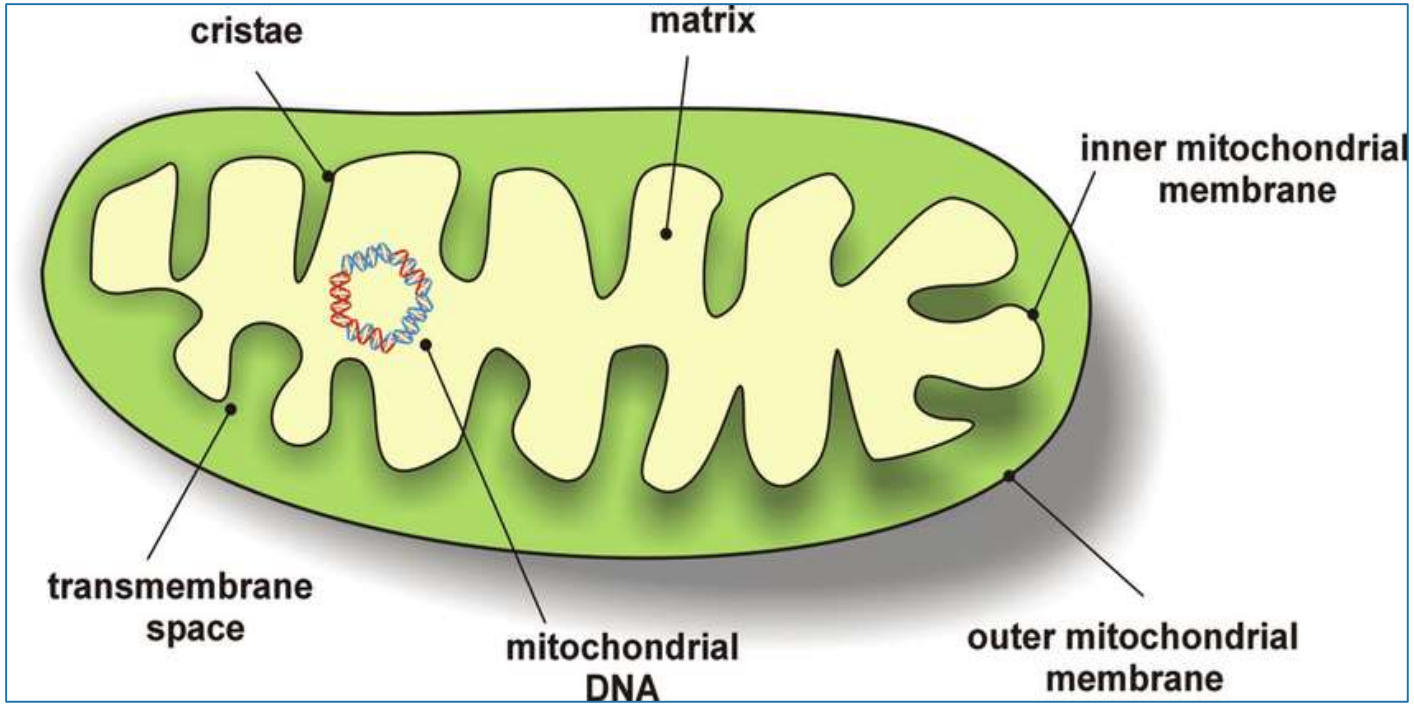
Mitochondria

- **Double Membrane:** Mitochondria have a double membrane structure: an outer mitochondrial membrane and an inner mitochondrial membrane, which create two distinct compartments within the organelle.

- **Cristae:** The inner mitochondrial membrane contains numerous infoldings called cristae, which increase the surface area for various metabolic processes.

- **Matrix:** The innermost compartment, known as the matrix, contains enzymes, DNA, and ribosomes, allowing for the synthesis of some mitochondrial proteins.

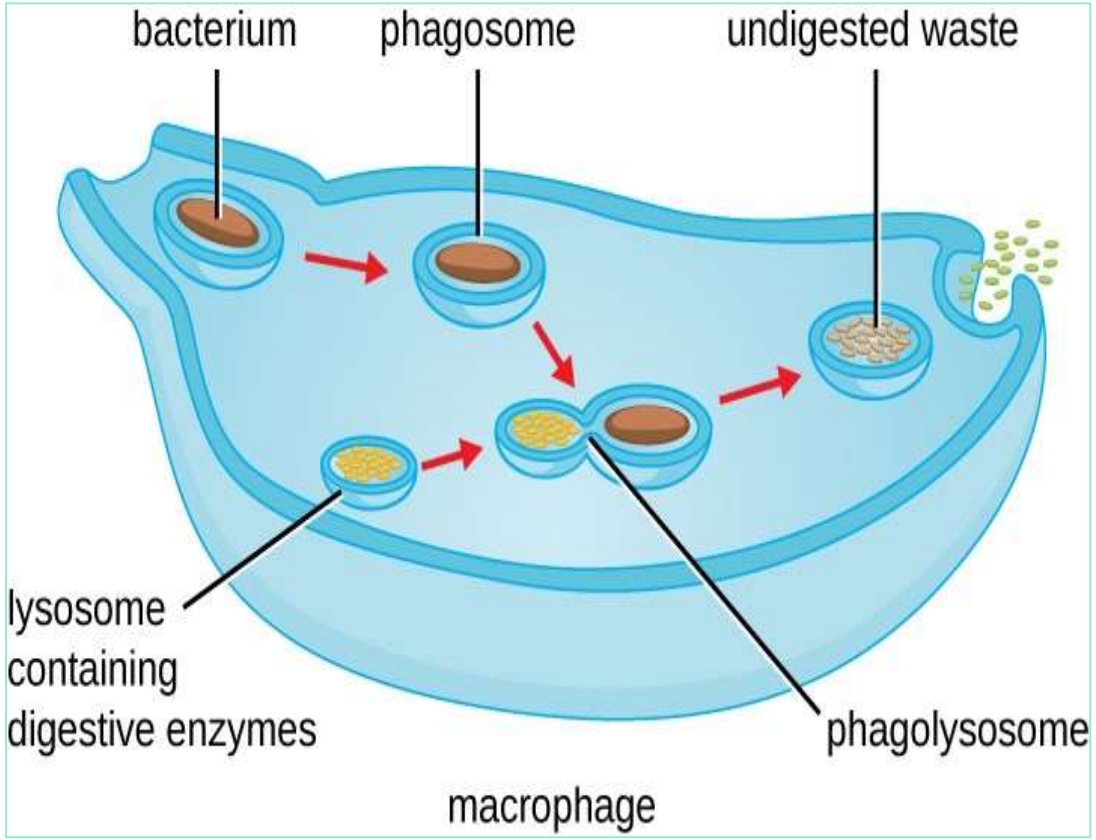
- **Mitochondrial DNA (mtDNA):** Mitochondria contain their own genetic material in the form of a small circular DNA molecule, separate from the cell's nuclear DNA.



Mitochondria are often referred to as the "powerhouses" of the cell because they are responsible for generating the majority of the cell's adenosine triphosphate (ATP), which is the primary source of cellular energy.

Structure of Cells (Cytoplasm)

Lysosomes



Lysosomes have hydrolytic enzymes that destroy pathogens and worn-out cells, so known as scavengers of the cells.

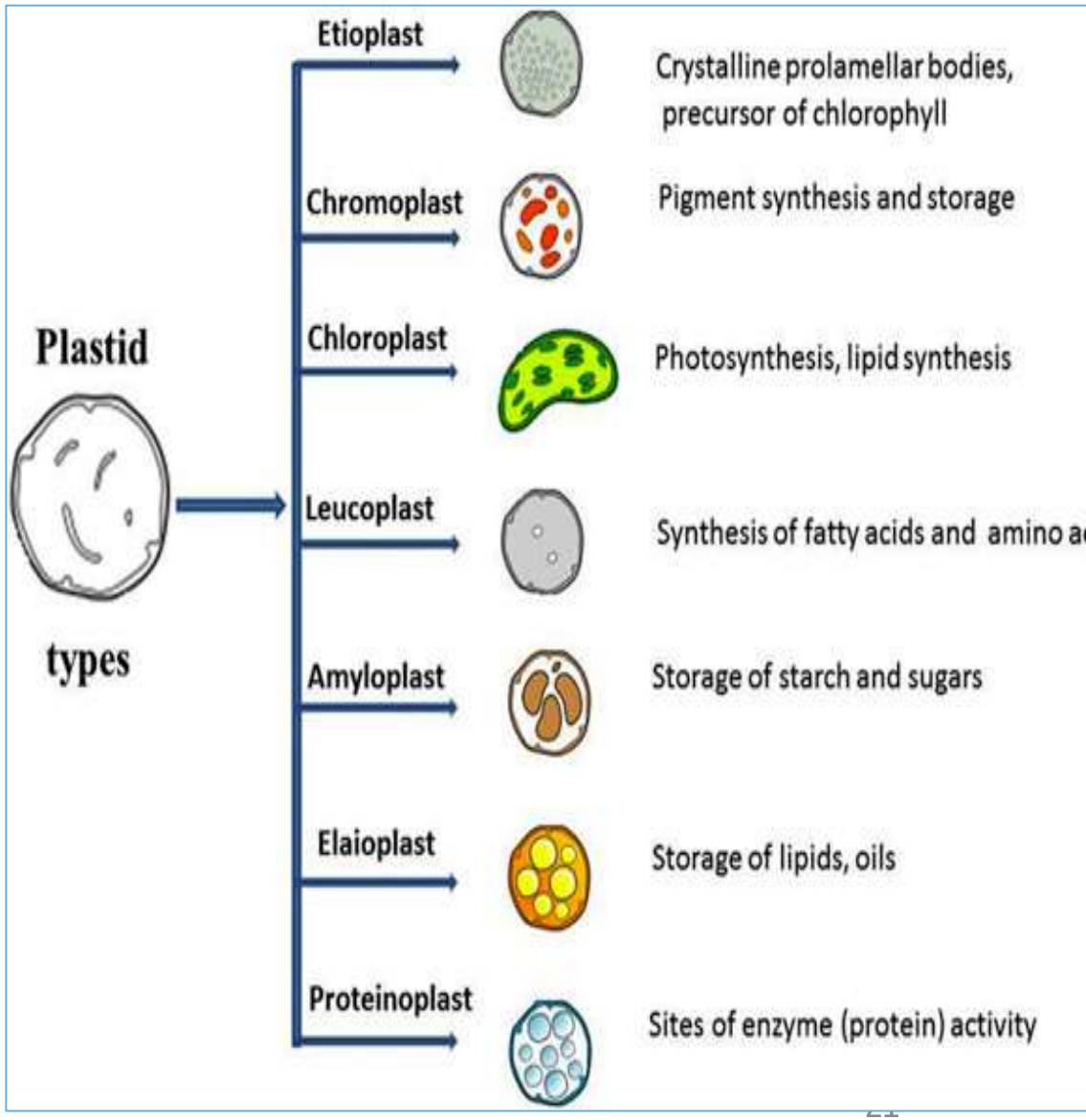
If something burst, the lysosomes release digestive enzymes with digests all the cells. This leads to the death of cells. Hence, Lysosomes are referred to as "suicide bags of cell".

Structure of Cells

Plastids

- **Chloroplasts** (Green coloured pigments)
- **Chromoplasts** (Different coloured pigments)
- **Leucoplasts** (Colourless pigments)

Plant cells ✓ animal cells X

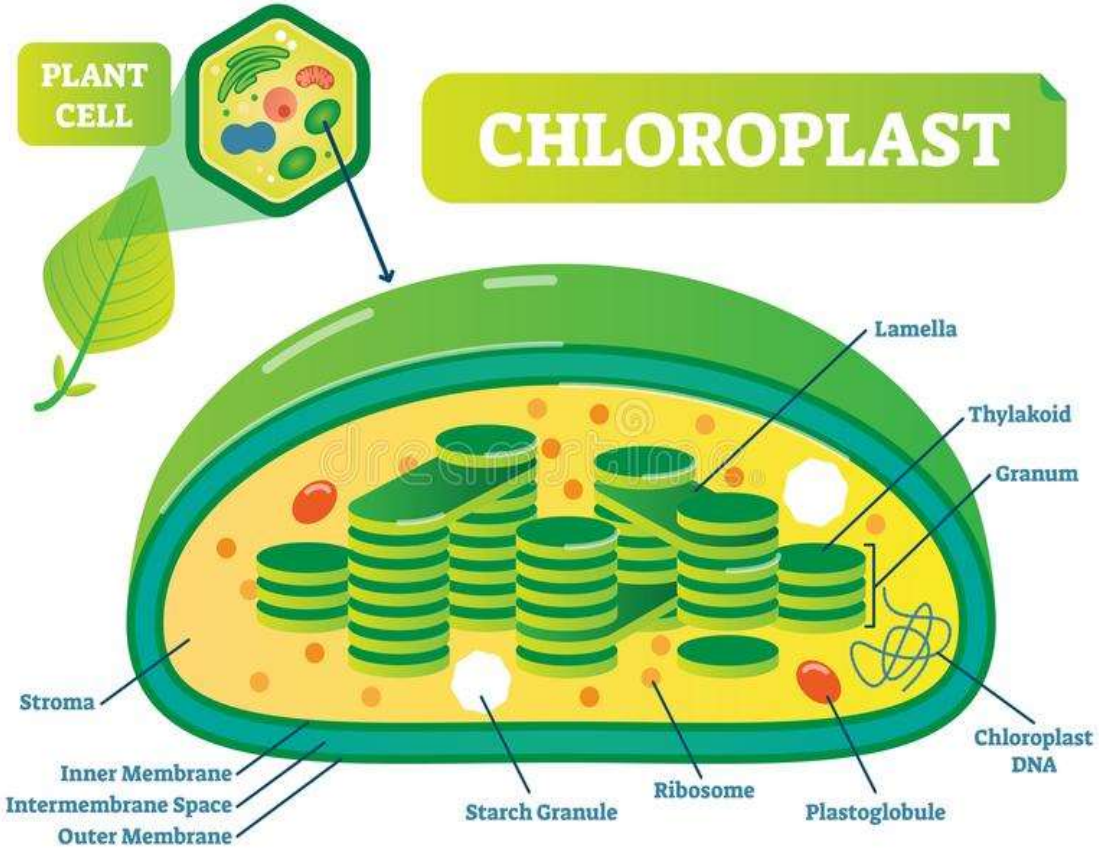


Structure of Cells
Chloroplast

- Chloroplast is an organelle that contains the photosynthetic pigment chlorophyll that captures sunlight and converts it into useful energy, thereby, releasing oxygen from water.

- These are found in mesophyll cells located in the leaves of the plants

- Chloroplast has its own extra-nuclear DNA and therefore are semiautonomous, like mitochondria. They also produce proteins and lipids required for the production of chloroplast membrane.



Structure of Cells
Chloroplast

Membrane Envelope
It comprises inner and outer lipid bilayer membranes. The inner membrane separates the stroma from the intermembrane space.

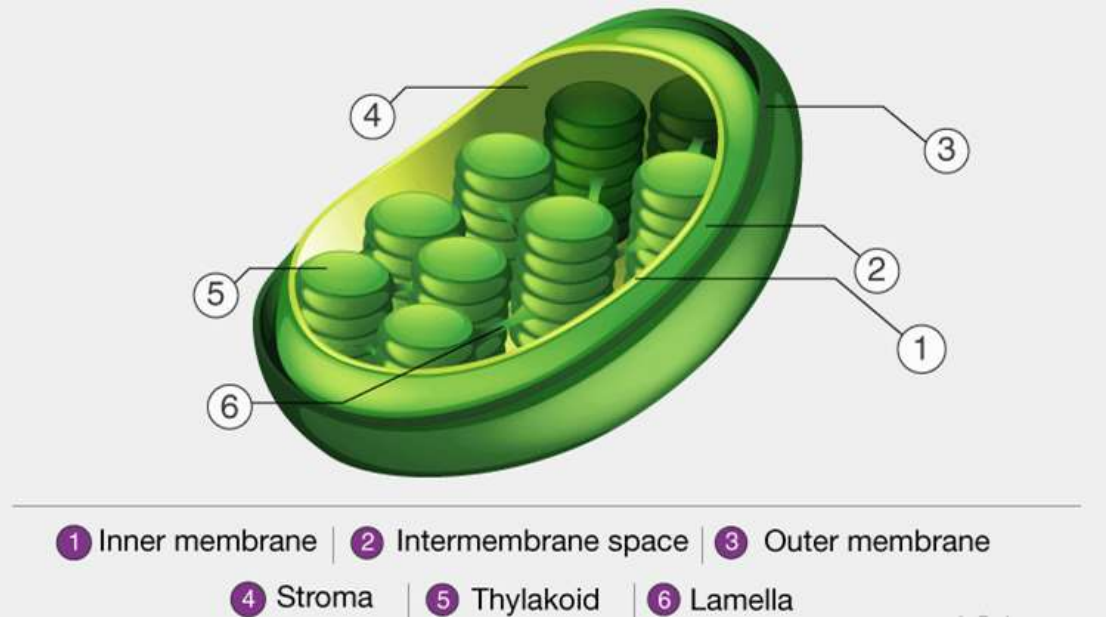
Intermembrane Space
The space between inner and outer membranes.

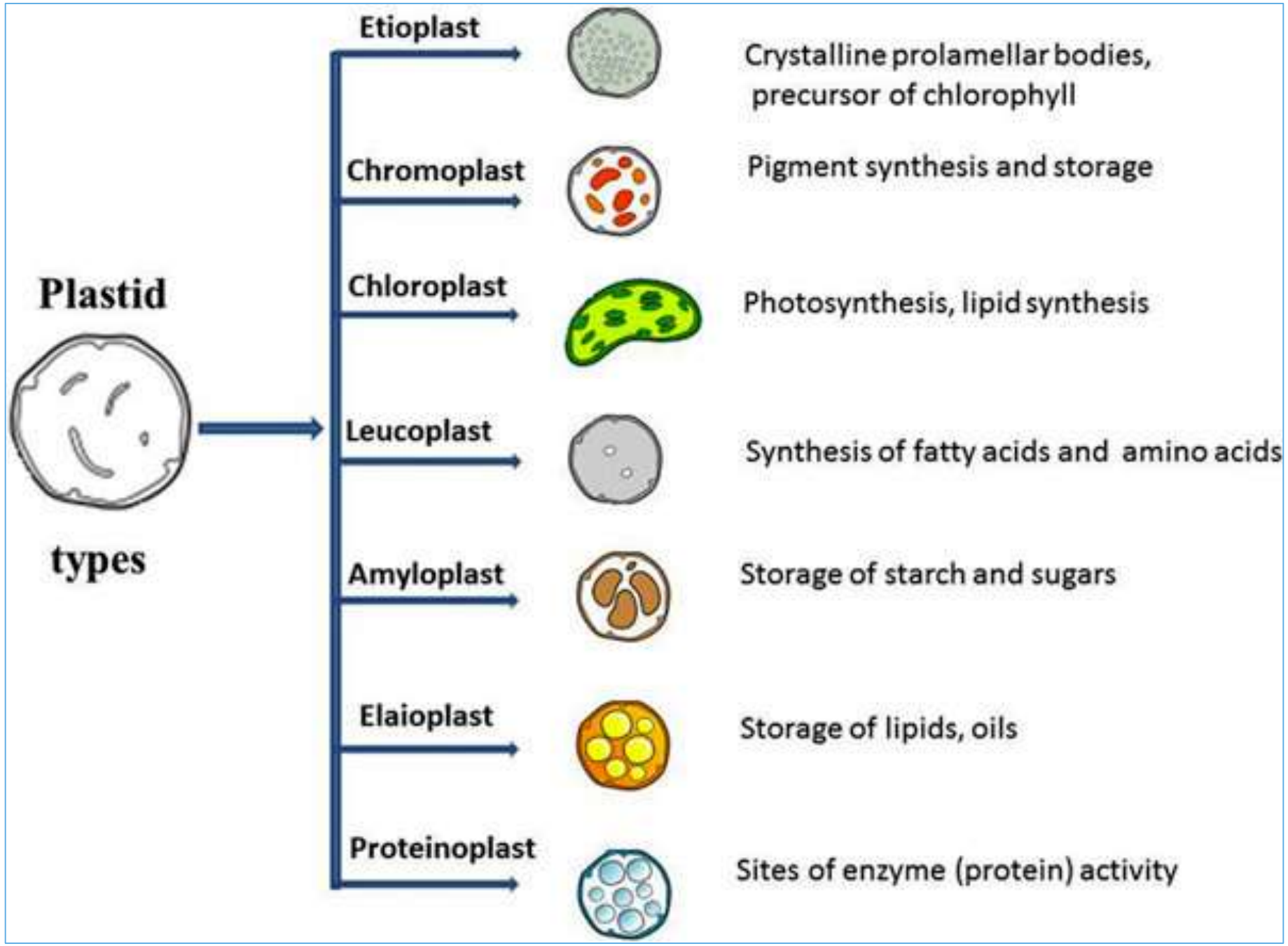
Thylakoid System (Lamellae)
The system is suspended in the stroma. It is a collection of membranous sacs called thylakoids or lamellae. The green coloured pigments called chlorophyll are found in the thylakoid membranes.

Stroma: It is a colourless, alkaline, aqueous, protein-rich fluid present within the inner membrane of the chloroplast present surrounding the grana.

Grana: Stack of lamellae in plastids is known as grana. These are the sites of conversion of light energy into chemical energy.

Chlorophyll: It is a green photosynthetic pigment that helps in the process of photosynthesis.

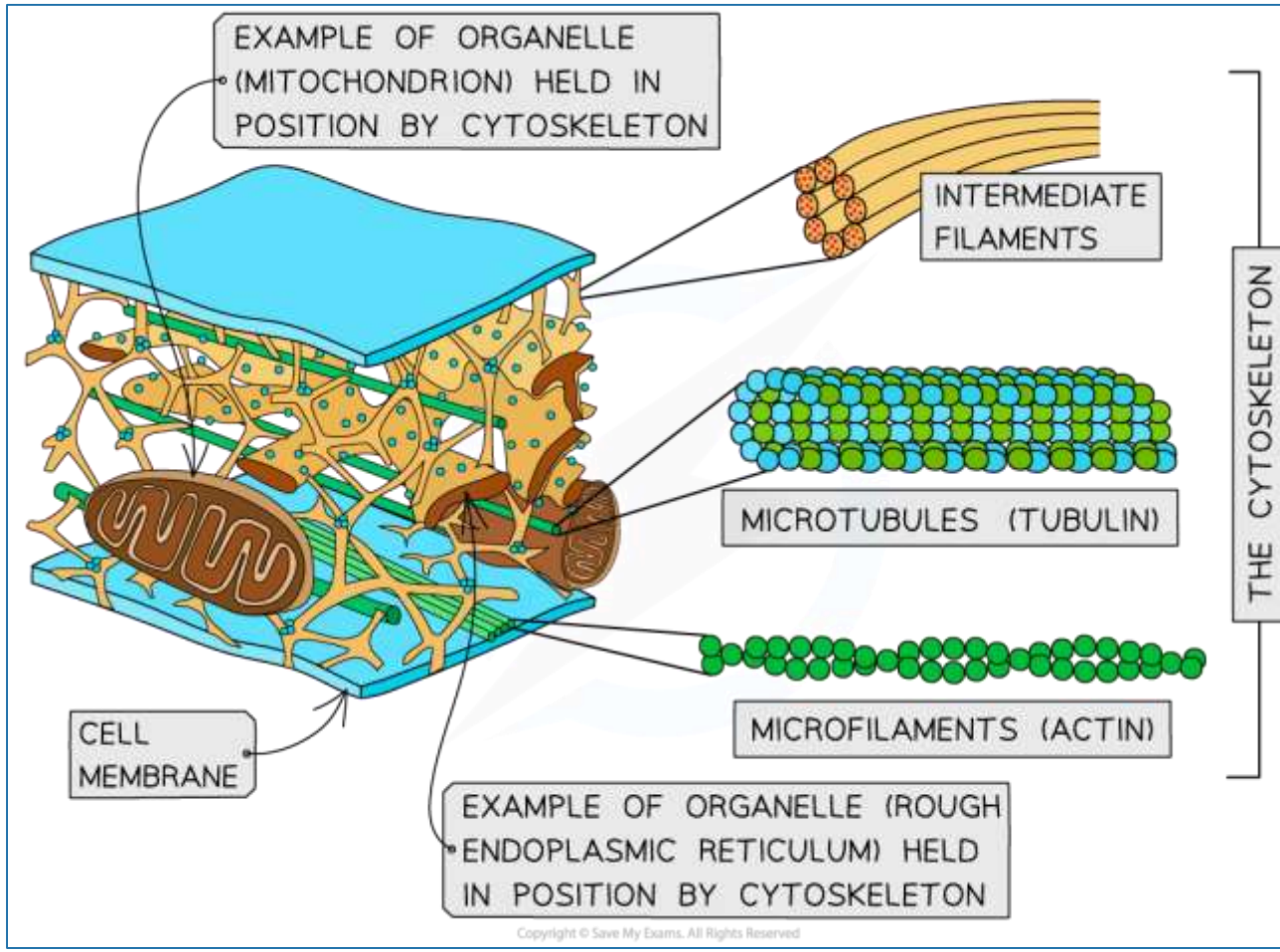



Plastid
types
Etioplast
Crystalline prolamellar bodies, precursor of chlorophyll
Chromoplast
Pigment synthesis and storage
Chloroplast
Photosynthesis, lipid synthesis
Leucoplast
Synthesis of fatty acids and amino acids
Amyloplast
Storage of starch and sugars
Elaioplast
Storage of lipids, oils
Proteinoplast
Sites of enzyme (protein) activity

Structure of Cells (Cytoplasm)

Cytoskeleton

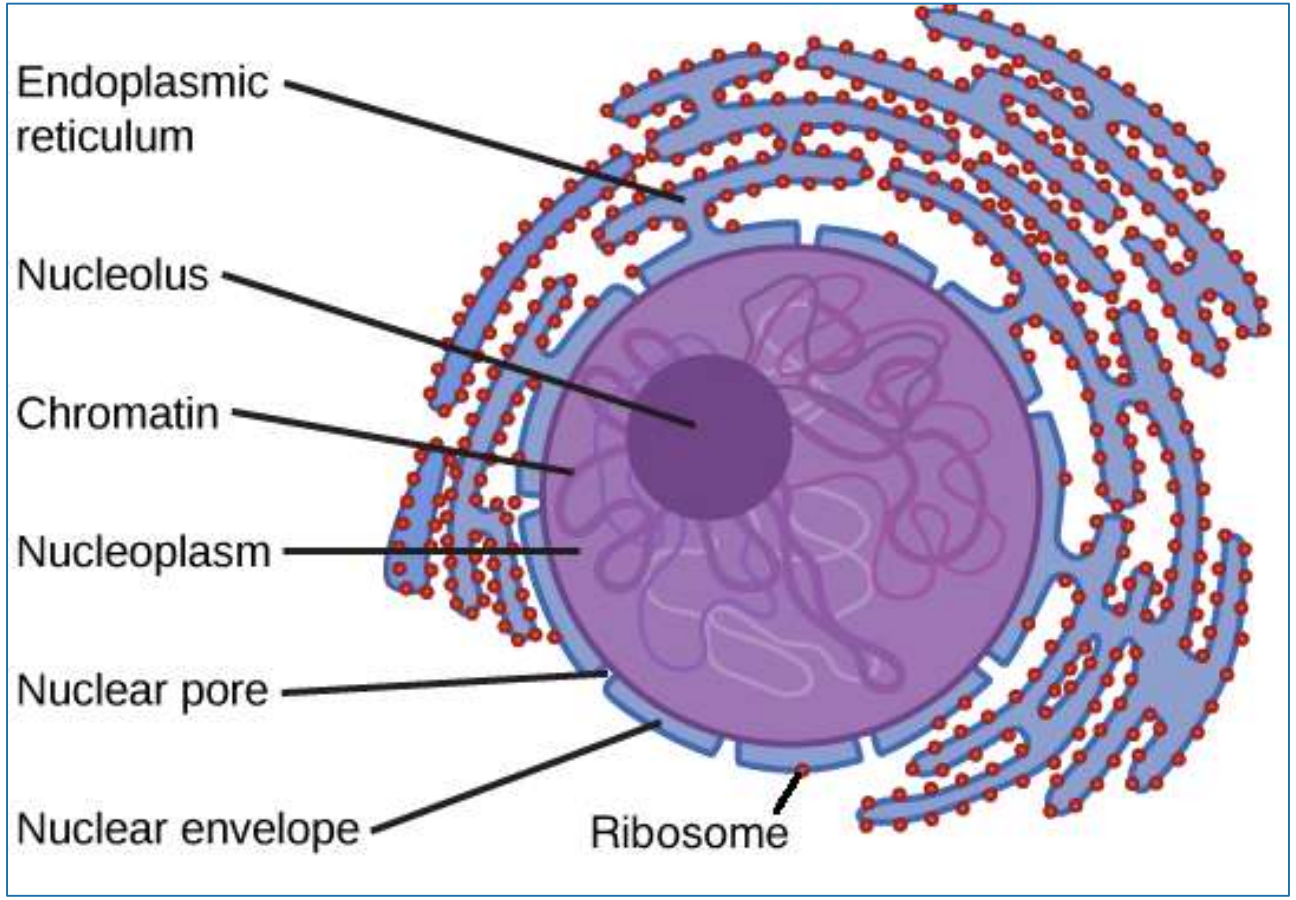
- Present in all eukaryotic cells
- Basis for cell shape and internal organization
- Allows organelle movement within cells and, in some cases, cell motility
    - Microtubules
    - Microfilaments
    - Intermediate Filaments



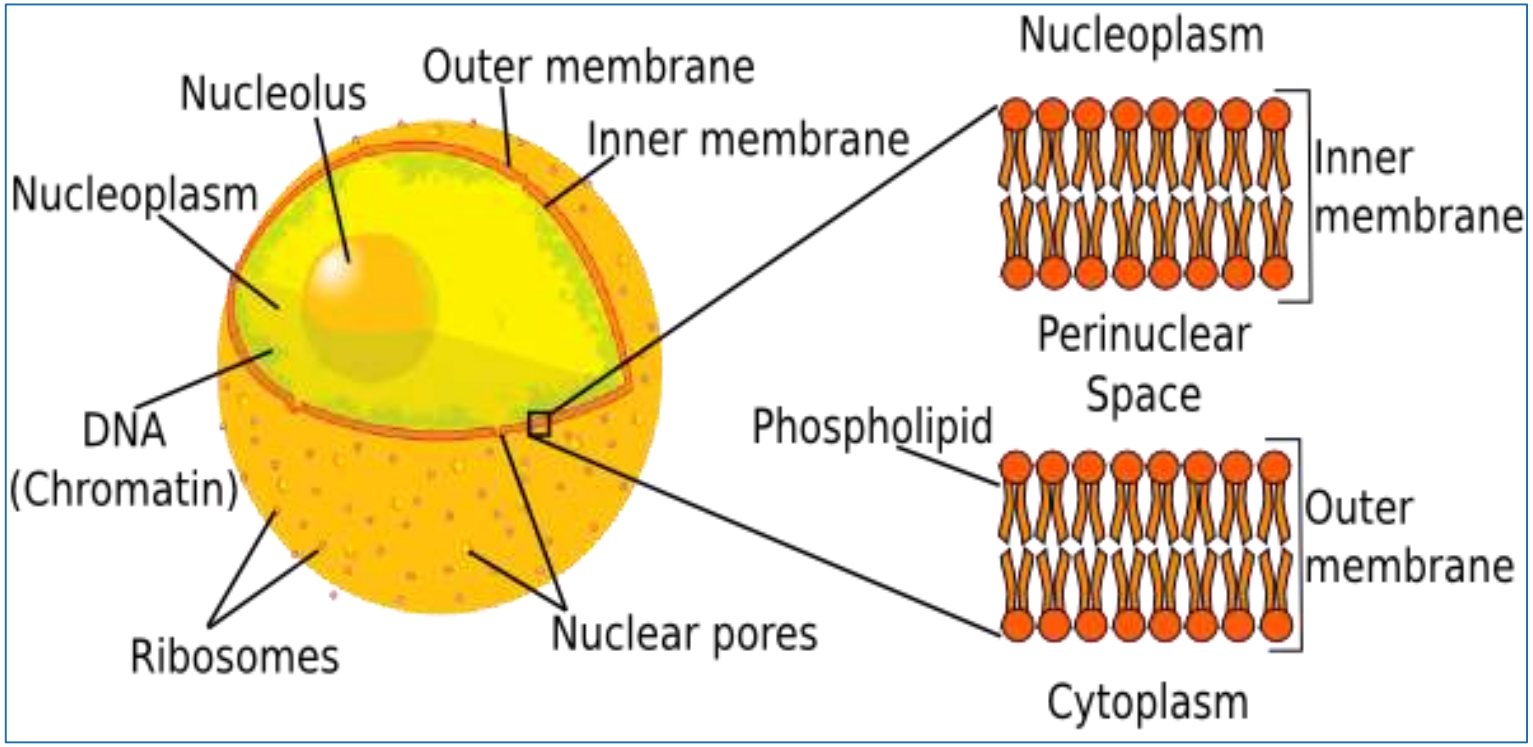
Structure of Cells
(Nucleus)

Components of Nucleus

- Nuclear envelope
- Nucleolus
- Nucleoplasm
- Nuclear pore
- Chromatin
- Chromosome/DNA



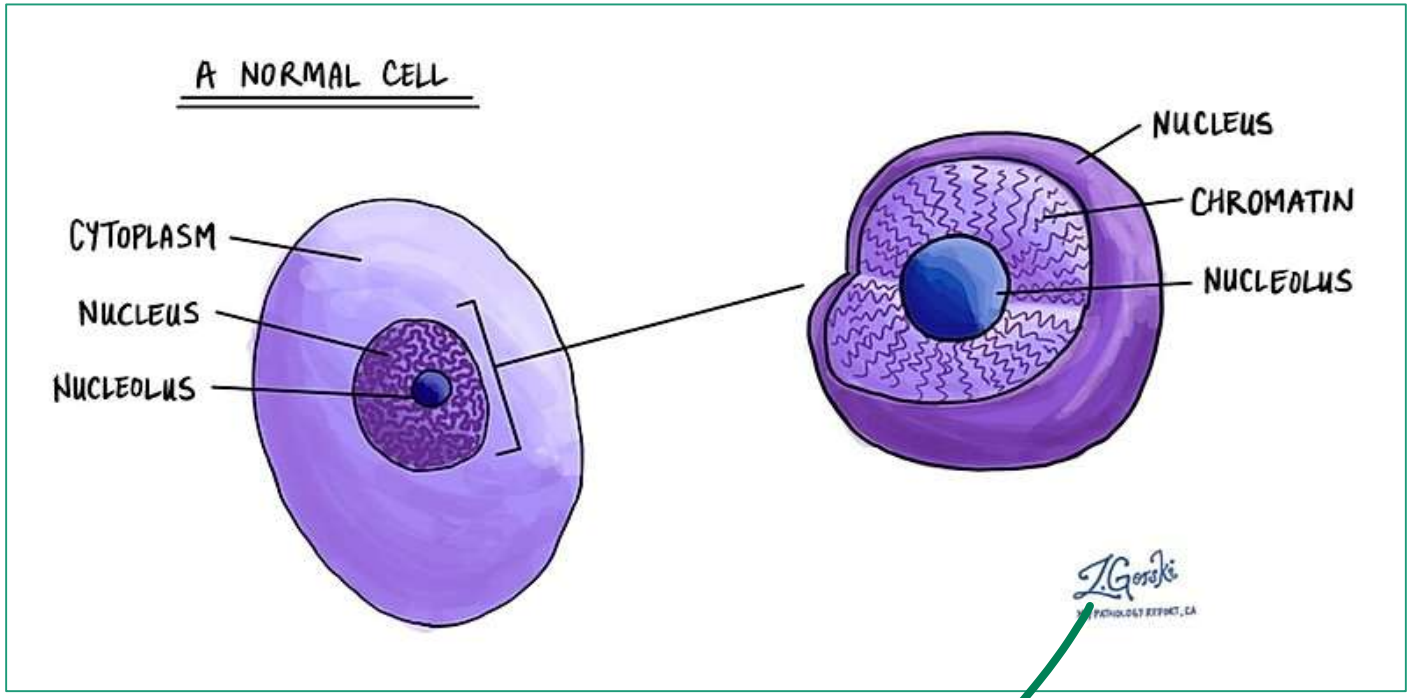
Structure of Cells
(Nucleus)



Nuclear Envelope

- Two outer membranes (lipid bilayers)
- Innermost surface has DNA attachment sites
- Pores span bilayer

Structure of Cells
(Nucleus)



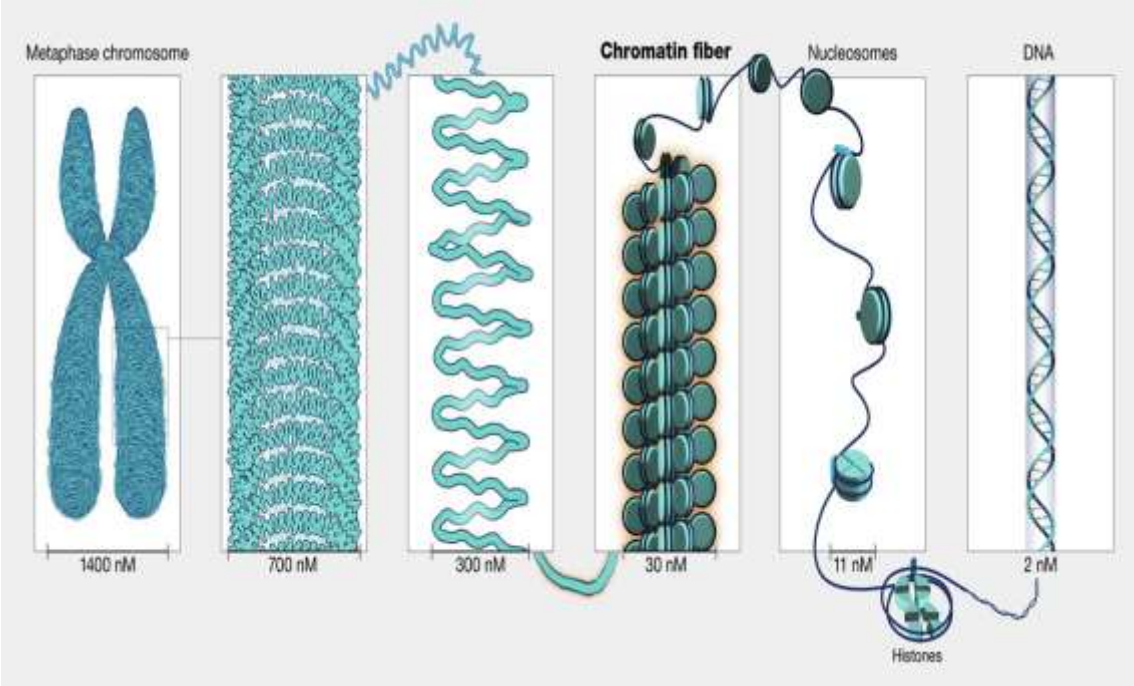
Nucleolus

- May be one or more
- Cluster of DNA and proteins
- Dense mass of material in nucleus
- Materials from which ribosomal subunits are built
- Subunits must pass through nuclear pores to reach cytoplasm

Structure of Cells
(Nucleus)

- **Nuclear Pores:** Tiny channels or pores in the nuclear envelope allow for the exchange of molecules between the nucleus and the cytoplasm.

- **Chromatin & Chromosomes:** The DNA is packaged by special proteins called histones to form chromatin. The chromatin further condenses to form chromosomes. This means chromatin is a lower order of DNA organization, whereas chromosomes are a higher order of DNA organization.



Structure of Cells
(Nucleus)

Functions of Nucleus

- Controls all cellular functions.
- Controls the synthesis of structural proteins.
- Contains genetic information for reproduction, development and behavior.
- Makes it easier to organize DNA and to copy it before parent cells divide into daughter cells

Function of cell organelles:

Organelle:	Function:
Nucleus	Contains DNA of the cell. Brain of the cell.
Nucleolus	Inner part of the nucleus that begins the assembly of ribosomes.
Chloroplast	Captures energy from light for photosynthesis. Solar panels of the cell.
Golgi Body/Apparatus	Modify, sort, and ship proteins and lipids. Post office of the cell.
Mitochondria	Makes ATP (energy). Powerhouse of the cell.
Ribosome	Creates polypeptide chains and proteins. Factory of the cell.
Smooth Endoplasmic Reticulum	Creates lipids for the cellular membrane.
Rough Endoplasmic Reticulum	Proteins are assembled and transported into/out of the cell.
Lysosome	Uses enzymes to break down old organelles and macromolecules. Garbage disposal of the cell.
Vacuole	Stores water, salts, and carbohydrates. Kitchen of the cell.
Cytoskeleton	Helps the cell maintain its shape; also involved in movement.
Cytoplasm	Material inside the cell membrane but outside of the nucleus.
Centriole	Help organize cell division.
Cell Wall	Gives plant cells extra stability and structure.
Cell Membrane	Controls what comes in and goes out of the cell.

THANK YOU